Kartik Sharma

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PROFESSIONAL SUMMARY

Computer Science Engineering student skilled in Python, Java, SQL, OOP with hands-on experience building Generative AI, LLM and RAG-based applications. Strong foundation in software development and problem solving, with a focus on developing practical AI solutions and learning emerging technologies.

EDUCATION

Vellore Institute Of Technology(VIT) Sept 2023 - Sept 2027
B.Tech in Computer Science Engineering (AI & ML)
CGPA: 8.03 / 10.0

TECHNICAL SKILLS

- **Languages:** Java, SQL, Python
- **Web& Backend:** HTML, CSS, JavaScript, Flask, REST APIs
- **AI/ML:** LLMs, RAG, Langchain, NLP, Scikit-learn, Sentence Transformers, Embeddings, Semantic Search
- **Frameworks & Libraries:** Streamlit, FAISS, NumPy, Pandas
- **Core Cs:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)
- **Database:** MySQL
- **Version Control:** Git, GitHub

EXPERIENCE

Google Cloud Generative AI Virtual Internship (SmartBridge) Nov 2025 - Jan 2026

- Explored Generative AI concepts and Google Cloud Vertex AI through hands-on activities focused on practical AI applications.
- Practiced prompt engineering by designing, refining and testing prompts to improve the relevance and quality of generated responses.
- Experimented with different prompting techniques and evaluated model outputs to understand how prompt structure influences response quality.

PROJECTS

Chat_Summarizer – AI-Powered Question Answering System GitHub
Python, FAISS, Sentence Transformers, Groq API

- Developed a full-stack Retrieval-Augmented Generation (RAG) web application to accurately extract, chunk,and process text from user-uploaded PDF documents.
- Engineered a semantic search pipeline utilizing Sentence Transformers for embedding generation and a FAISS vector database, configuring the system to efficiently retrieve the top 2 most relevant document chunks.
- Integrated the Groq LLM API Key to generate context-grounded, markdown-formatted answers, providing users with exact source references and real-time document processing statistics.

ScriptSense — RAG-based Youtube Transcript Analyzer GitHub
Python, Streamlit, RAG, Groq API, LLMs, Sentence Transformers, FAISS

- Built a GenAI-powered Q&A application using RAG, enabling users to query long YouTube/Podcast transcripts using natural language.
- Implemented text chunking, Sentence Transformer embeddings and FAISS vector search to retrieve relevant contextual information.
- Developed the application using Python and Streamlit, with retry logic and model fallbacks for improved LLM API reliability.

CERTIFICATIONS

- Google IT Support Professional Certification
- Google AI Essentials Certification
- Introduction to GenAI- Udemy
- Applied Machine Learning in Python-University of Michigan
- Advanced Software Engineering & Development Program - MPOnline.gov